

$$-\left[\cancel{\bar{\sigma}_{zz}(x_I)} - \bar{\sigma}_{zz}(x_T)\right] - \left[\bar{\sigma}_{zz}(x_R) - \cancel{\bar{\sigma}_{zz}(x_I)}\right] = \Delta \text{GPE}^*$$

$$\text{trench pull} \equiv -(\bar{\sigma}_{zz}(x_I) - \bar{\sigma}_{zz}(x_T))$$

$$\text{ridge push} \equiv -(\bar{\sigma}_{zz}(x_R) - \bar{\sigma}_{zz}(x_I))$$

trench x_T

$$\tau_{zx} = 0$$

x_I (first $w = 0$)

$$\tau_{zx} = 0$$

ridge x_R

z_0

non-isostatic domain
 $\sim 10^2$ km



isostatic domain
a few $\times 10^3$ km

(not to scale)

$\bar{\sigma}_{xx}$ →

← $\bar{\sigma}_{xx}$

at x_T :
 $-\bar{\sigma}_{zz} \neq \bar{P}_L$

at x_R :
 $-\bar{\sigma}_{zz} = \bar{P}_L$
 $N_D \approx 0$

horizontal force balance:

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$$\Delta \bar{\sigma}_{xx} + F_B = 0$$

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$$\Rightarrow \Delta \text{GPE}^* \approx \Delta N_D$$

$$\Rightarrow N_D(x_I) \approx F_B - \Delta \text{GPE}^*$$

z_c

$$F_B = \int \tau_{zx} dx \approx 0$$

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